

Multiplexing

Whenever the bandwidth of a medium linking two devices is greater than the bandwidth needs of the devices, the link can be shared.

Multiplexing is the set of techniques that allow the simultaneous transmission of multiple signals across a single data link.

If the bandwidth of a link is greater than the bandwidth needs of the devices connected to it, the bandwidth is wasted.

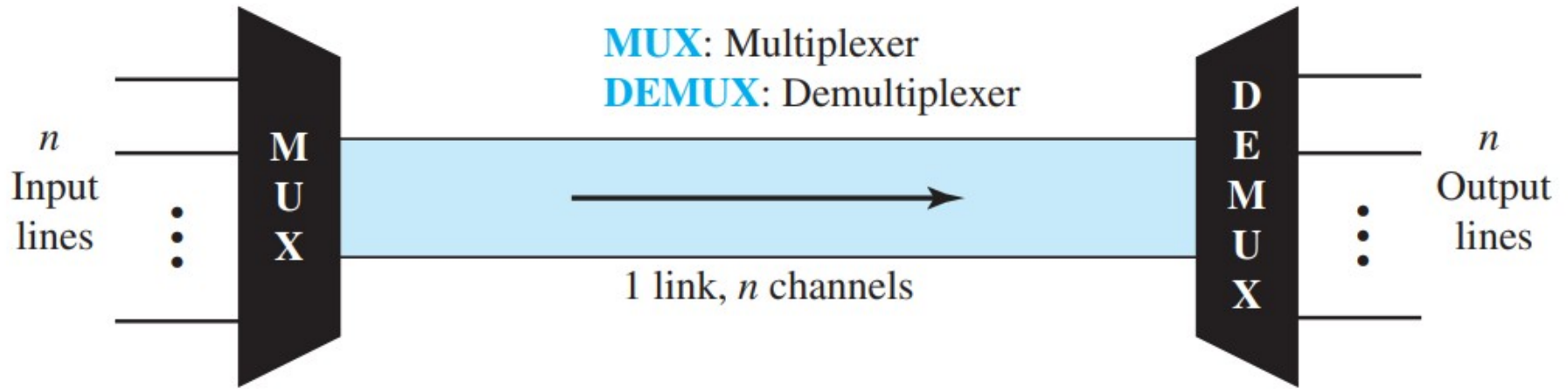
An efficient system maximizes the utilization of all resources; bandwidth is one of the most precious resources we have in data communications.

Uses of Multiplexing

Multiplexing is used for a variety of purposes in data communications to enhance the efficiency and capacity of networks. Here are some of the main uses:

- **Efficient Utilization of Resources:** Multiplexing allows multiple signals to share the same communication channel, making the most of the available bandwidth. This is especially important in environments where bandwidth is limited.
- **Telecommunications:** In telephone networks, multiplexing enables the simultaneous transmission of multiple telephone calls over a single line, enhancing the capacity of the network.
- **Internet and Data Networks:** Multiplexing is used in internet communications to transmit data from multiple users over a single network line, improving the efficiency and speed of data transfer.
- **Satellite Communications:** Multiplexing helps in efficiently utilizing the available bandwidth on satellite transponders, allowing multiple signals to be transmitted and received simultaneously.

Multiplexing

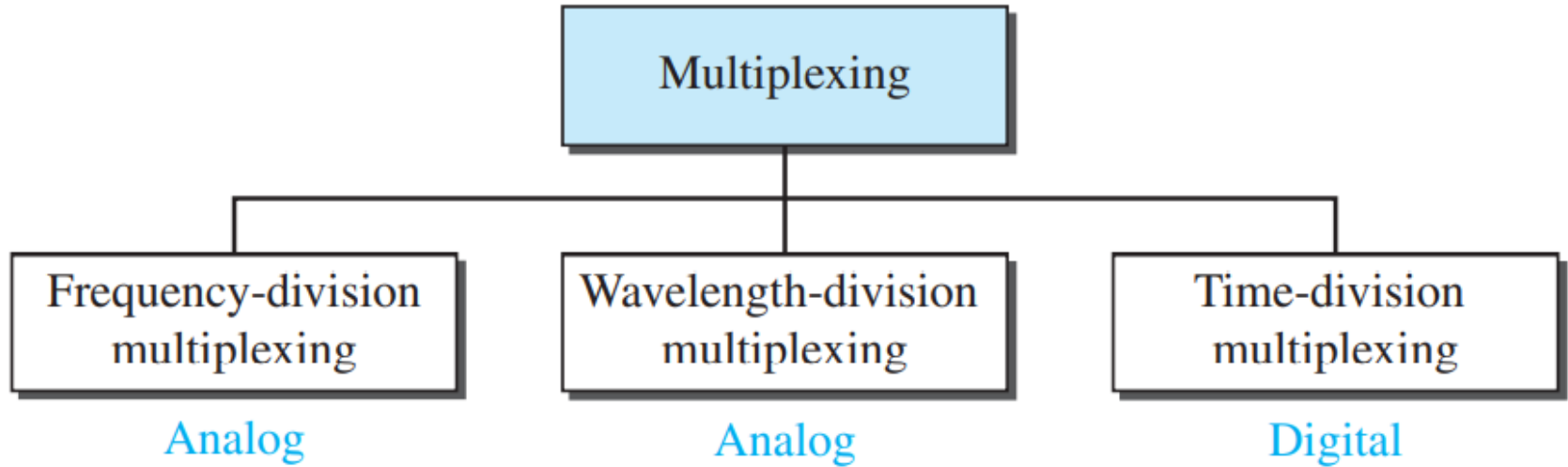


Multiplexing

There are three basic multiplexing techniques: ***frequency-division multiplexing, wavelength-division multiplexing, and time-division multiplexing.***

The first two are techniques designed for analog signals, the third, for digital signals

Multiplexing



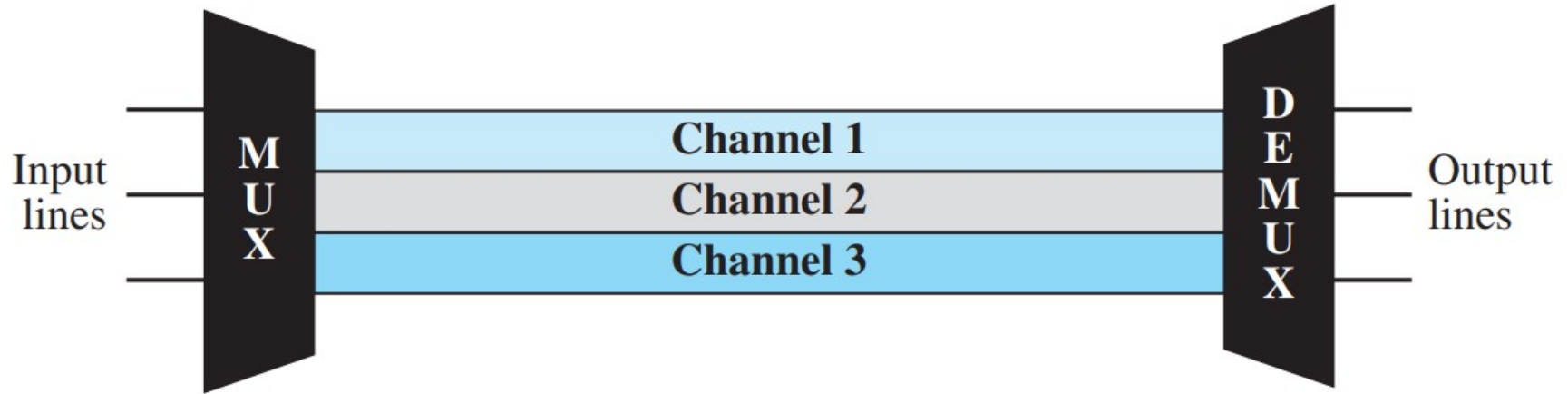
Frequency-Division Multiplexing

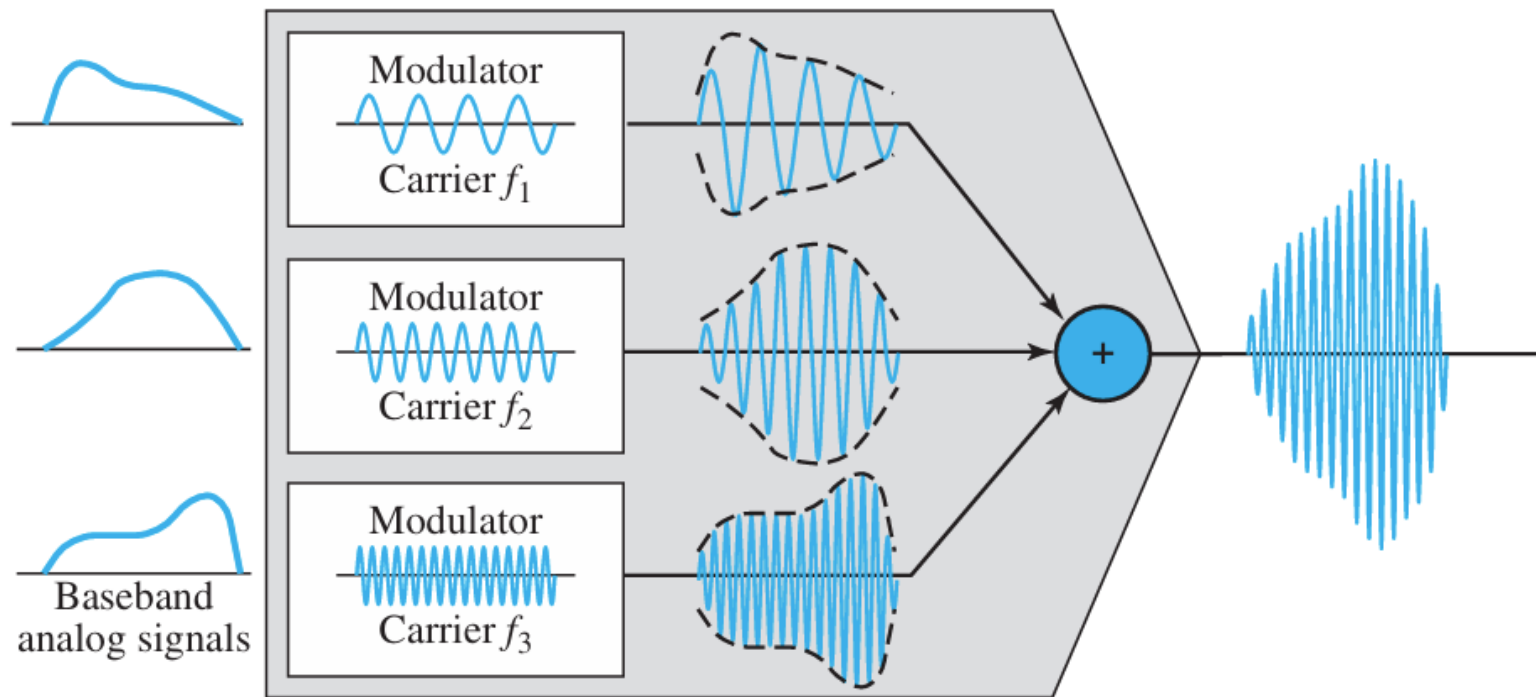
Frequency-division multiplexing (FDM) is an analog technique that can be applied when the bandwidth of a link (in hertz) is greater than the combined bandwidths of the signals to be transmitted.

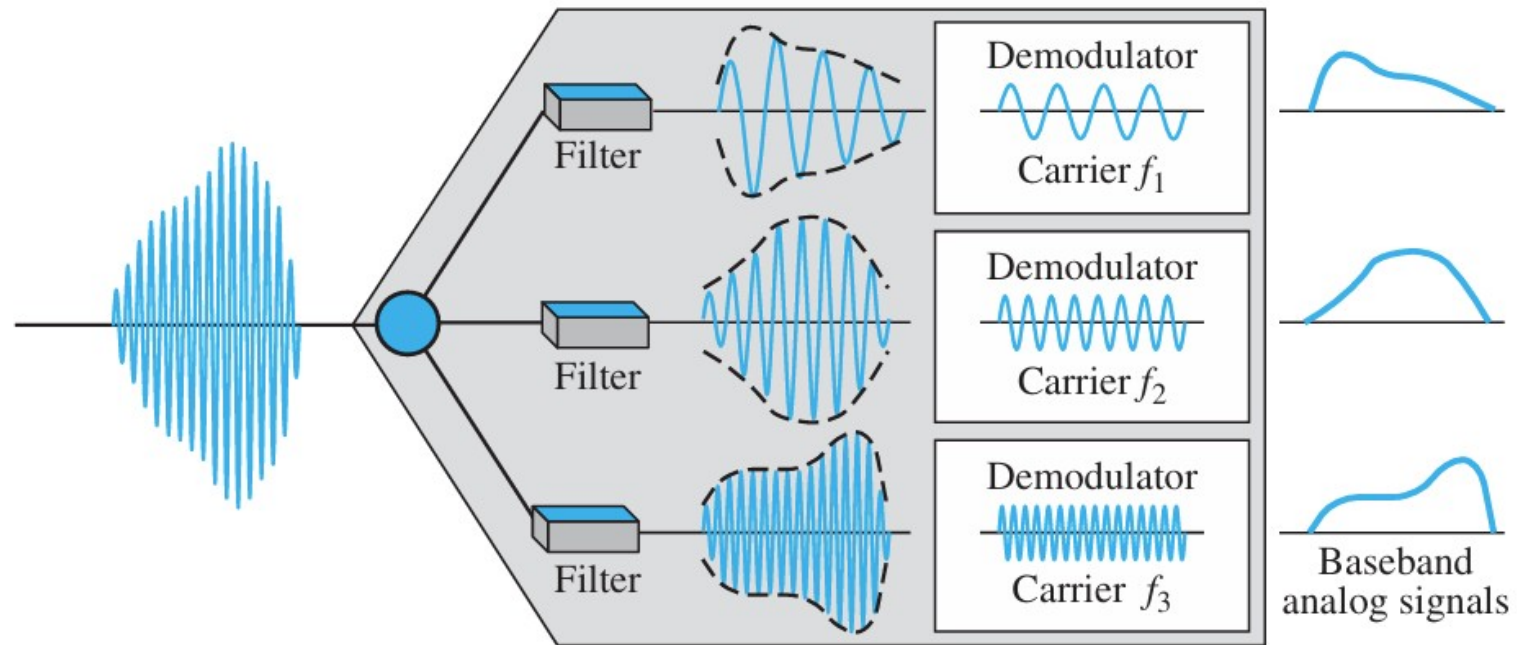
In FDM, signals generated by each sending device modulate different carrier frequencies.

These modulated signals are then combined into a single composite signal that can be transported by the link.

Frequency-Division Multiplexing







Frequency-Division Multiplexing

Carrier frequencies are separated by sufficient bandwidth to accommodate the modulated signal.

These bandwidth ranges are the channels through which the various signals travel.

Channels can be separated by strips of unused bandwidth—**guard bands**—to prevent signals from overlapping.

In addition, carrier frequencies must not interfere with the original data frequencies.

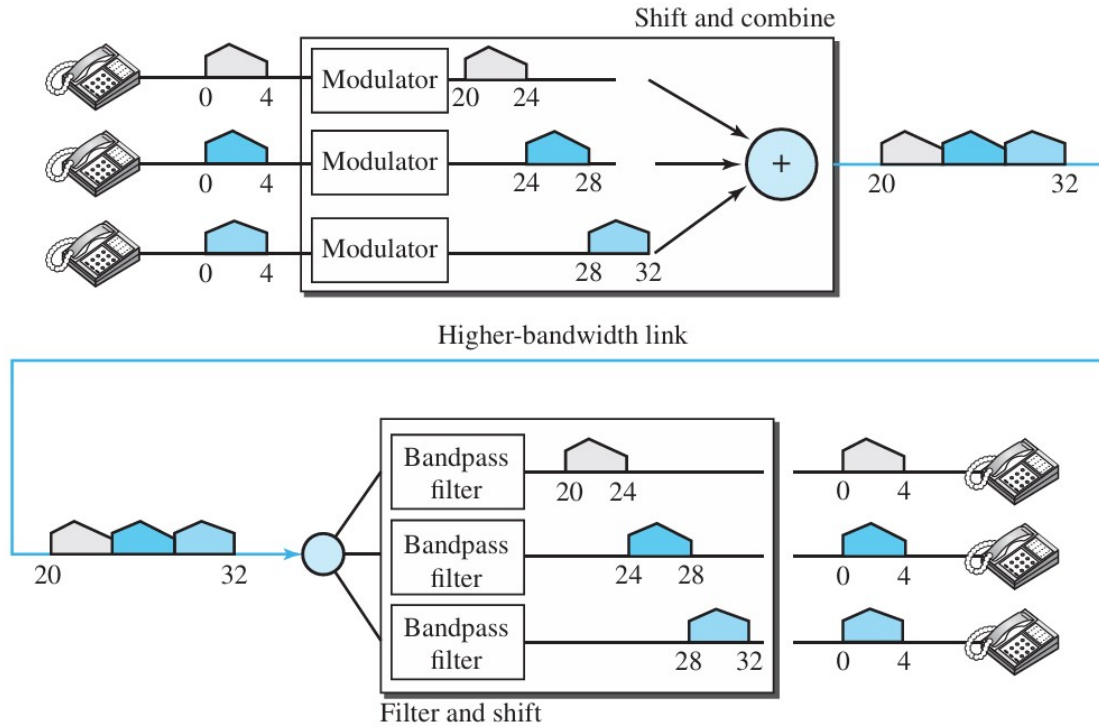
Note: *Frequency Division Multiplexing is used in radio and television transmission.*

Frequency-Division Multiplexing

Example:

Assume that a voice channel occupies a bandwidth of 4 kHz. We need to combine three voice channels into a link with a bandwidth of 12 kHz, from 20 to 32 kHz. Show the configuration, using the frequency domain. Assume there are no guard bands.

Frequency-Division Multiplexing

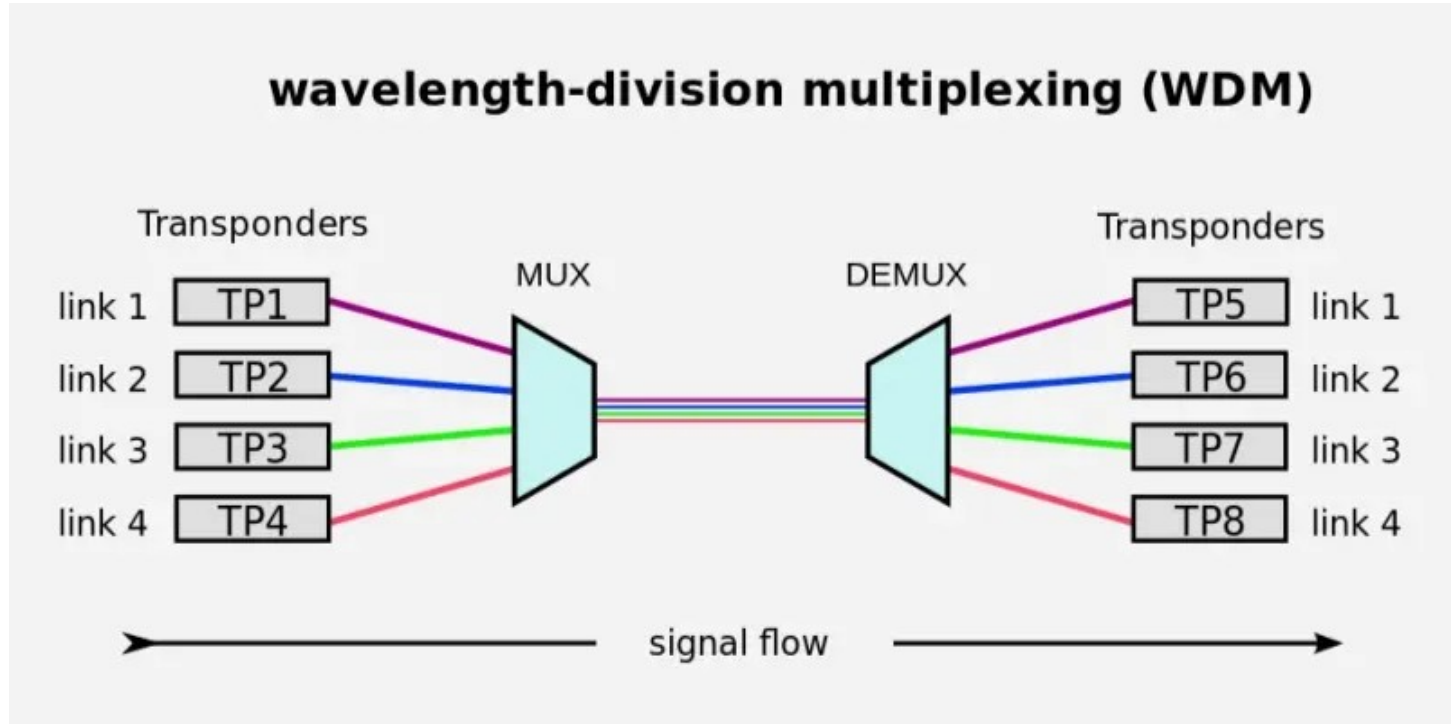


Wavelength-Division Multiplexing

Wavelength Division Multiplexing (WDM) is a multiplexing technology used to increase the capacity of optical fiber by transmitting multiple optical signals simultaneously over a single optical fiber, each with a different wavelength.

- Each signal is carried on a different wavelength of light, and the resulting signals are combined onto a single optical fiber for transmission.
- At the receiving end, the signals are separated by their wavelengths, demultiplexed and routed to their respective destinations.
- It is used in telecommunications, cable TV, ISPs, and data centers for high-speed, long-distance data transmission.

Wavelength-Division Multiplexing



Wavelength-Division Multiplexing

What a WDM Transponder Does

A **WDM transponder** is an optical-electrical-optical (O/E/O) converter that:

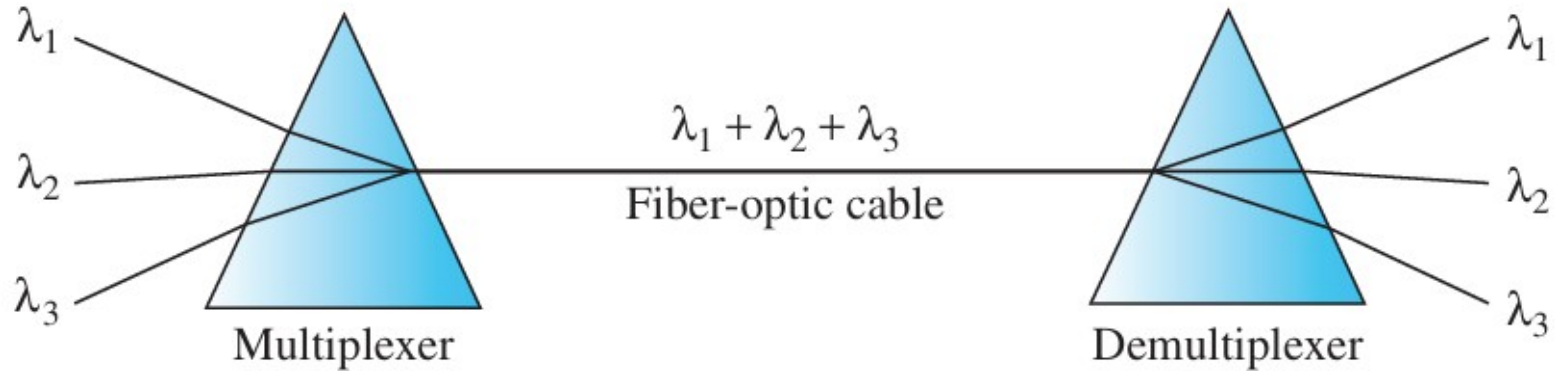
1. **Receives** an incoming optical signal (often from legacy equipment or a different wavelength).
2. **Converts** it to an electrical signal.
3. **Processes** it with 3R functions:
 - **Reshape** – cleans up signal distortions.
 - **Retime** – synchronizes the clock.
 - **Reamplify** – boosts the signal power
4. **Reconverts** it to an optical signal at a target WDM wavelength for multiplexing.

Wavelength-Division Multiplexing

Why Transponders Are Used in WDM

- **Wavelength conversion:** Bridges equipment using different wavelengths or standards
- **Interoperability:** Connects networks from different providers or with different fiber types (e.g., multimode to single-mode).
- **Signal regeneration:** Extends reach by cleaning and amplifying the signal
- **Flexibility:** Allows upgrading or reconfiguring network capacity without replacing the fiber infrastructure

Wavelength-Division Multiplexing



Wavelength-Division Multiplexing

- Although WDM technology is very complex, the basic idea is very simple. We want to combine multiple light sources into one single light at the multiplexer and do the reverse at the demultiplexer.
- The combining and splitting of light sources are easily handled by a prism.
- A prism bends a beam of light based on the angle of incidence and the frequency.
- Using this technique, a multiplexer can be made to combine several input beams of light, each containing a narrow band of frequencies, into one output beam of a wider band of frequencies.
- A demultiplexer can also be made to reverse the process.

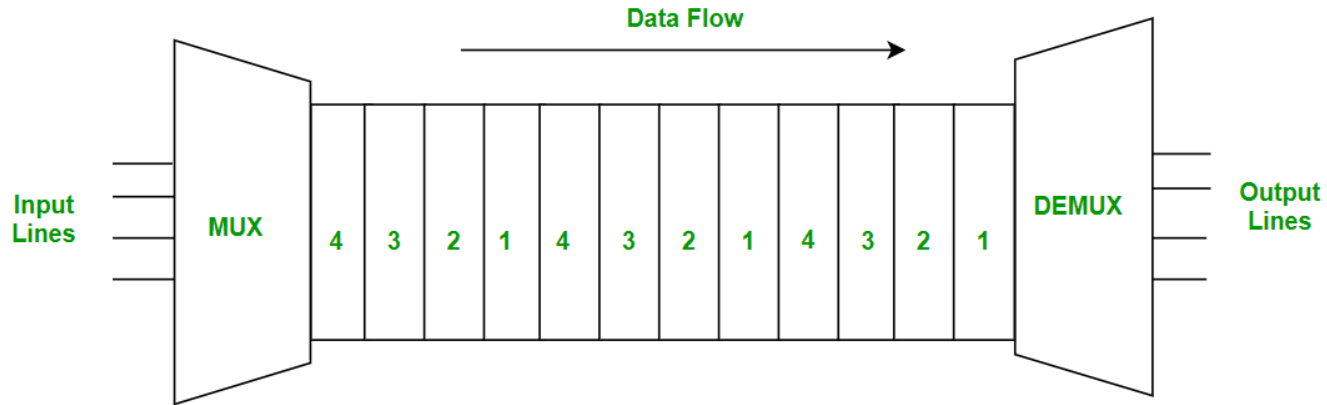
Time-Division Multiplexing

Time-division multiplexing is multiplexing wherein FDM, instead of sharing a portion of the bandwidth in the form of channels, in TDM, time is shared.

Each connection occupies a portion of time in the link.

In Time Division Multiplexing, all signals operate with the same frequency (bandwidth) at different times.

Time-Division Multiplexing



Time-Division Multiplexing

Working Principle

1. **Signal Segmentation:** Incoming signals from multiple sources are divided into small, fixed-length segments or units.
2. **Time Slot Assignment:** Each segment is assigned a specific time slot within a repeating frame.
3. **Multiplexing:** A multiplexer combines these segments into a single composite signal, transmitting them sequentially over the shared channel.
4. **Demultiplexing:** At the receiver, a demultiplexer separates the composite signal back into the original signals using synchronized timing

Time-Division Multiplexing

Types of TDM

1. Synchronous TDM:

- Time slots are pre-assigned and fixed for each input, even if a source has no data to send. This is commonly used in digitized voice streams.

2. Asynchronous (Statistical) TDM:

- Time slots are dynamically allocated based on the availability of data from each source, improving channel efficiency by avoiding empty slots

Advantages of Multiplexing

- **Efficient Use of Bandwidth:** You can send more than one signal over a single channel. This way, you can use the channel's capacity more efficiently.
- **Increased Data Transmission:** Multiplexing can significantly boost the amount of data that can be sent over a network simultaneously, enhancing overall transmission capacity.
- **Scalability:** Multiplexing allows networks to easily expand and accommodate more data streams without requiring significant changes to the existing infrastructure.
- **Flexibility:** Different types of multiplexing (TDM, FDM, WDM) can be used based on the specific needs and characteristics of the communication system, providing flexibility in network design.